

Architecture Overview

Overtone - Meeting Join & Live Grounding

Audience: Technical partnership discussion

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Status: Active development - demo / architecture stage

Classification: Confidential - partner review

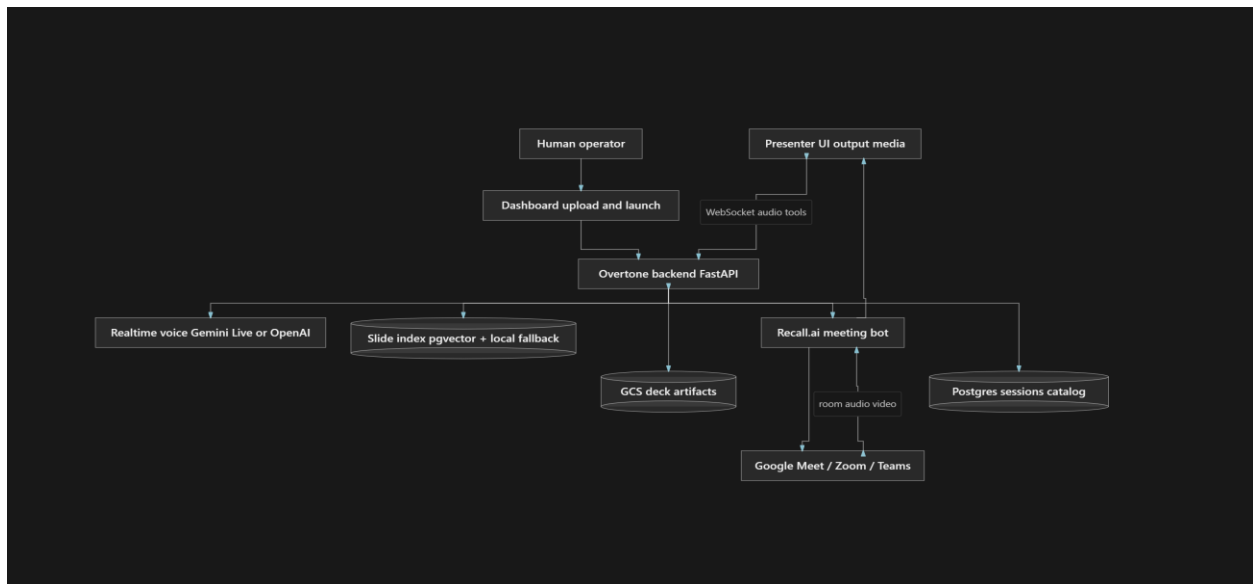
1. Framing

Overtone is a **live meeting layer**: a bot joins Google Meet, Zoom, or Teams as a participant, presents a deck, listens with realtime speech-to-speech, and answers from **indexed session sources** - with a design goal to abstain or escalate when grounding is weak (not invent).

It is **not** a finished “zero-hallucination guarantee” product. The honest framing is: traceable, source-bound outputs plus a hard stop when the source cannot support an answer.

2. System context

Operator uses the Dashboard to upload decks and launch bots. The Backend (FastAPI) creates a Recall.ai bot that joins the meeting room and opens the Presenter UI as the bot camera. The Presenter connects to the Backend over WebSocket for audio and tools. The Backend bridges to realtime voice (Gemini Live preferred, OpenAI Realtime as fallback) and retrieves grounded slide text from pgvector, with artifacts in GCS and session state in Postgres.



Component	Role
Dashboard	Upload decks, configure agents, launch bots into meetings
Recall.ai	Meeting participant join; virtual camera / audio path
Presenter UI	Deck display as bot camera; mic capture; play spoken audio
Backend relay	Session state, tool execution, voice bridge
Realtime voice	Speech-to-speech + function / tool calling
pgvector + Vision	Per-page extract, embed, retrieve grounded slide text
GCS + Postgres	Artifacts and durable session catalog

3. Session lifecycle

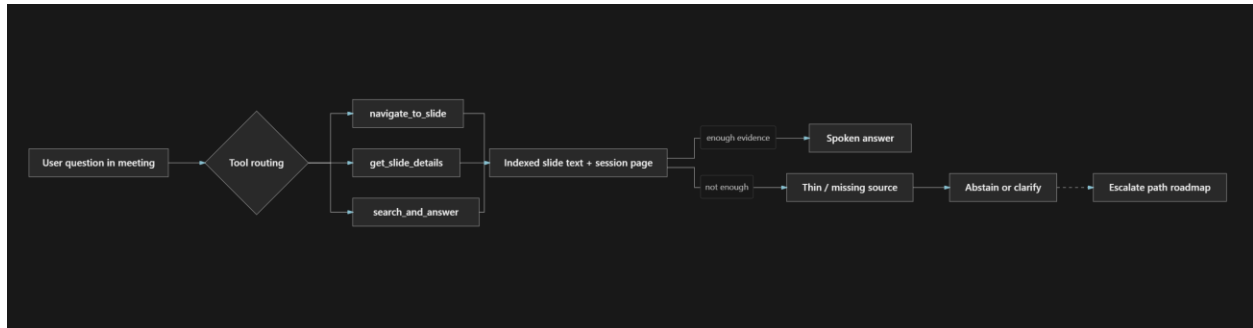
- Upload PDF / PPTX → store → Vision extract per page → quality gate → embeddings → pgvector → status ready
- Launch with meeting URL + knowledge base → create session → Recall bot with output-media presenter URL
- Bot joins room → Presenter WebSocket → Live voice session
- Participant audio → relay → model; model tool calls → navigate / get_slide_details / search → spoken answer from tool results only

4. Platform layers

Layer	Function	Current technology
Capture / join	Enter call as participant; camera shows slides	Recall.ai
Presenter	Deck UI, mic, bot audio playback	React (output-media page)
Realtime voice	Speech-to-speech + tools	Gemini Live / OpenAI Realtime
Orchestration	State, tools, mute, page position	FastAPI relay + executor
Grounding	Per-slide retrieval	Vision → embeddings → pgvector
Persistence	Files, catalog, session extras	GCS + Postgres

5. Grounding path

User question → tool routing (navigate / get_slide_details / search) → indexed slide text + session page.
Enough evidence → spoken answer. Insufficient evidence → abstain / clarify → escalate path (roadmap).



Design principles (escalate-don't-guess)

- Prefer the currently visible slide for “explain this / what’s on this.”
- Search the deck index before any external lookup.
- If retrieval is empty or stub-thin → do not invent; clarify or state not in source.
- Human escalate (expert as ground truth) is the intended next architectural step — not claimed as fully shipped today.

6. Indexing pipeline

Upload → store (GCS / local) → render page images → Vision per page → title / body / searchable_content → quality gate (reject stub pages) → embeddings → pgvector → presentation status ready. Vision stubs such as title-only “Slide 4” must not be marked ready.

7. Realtime tools

Tool	Role
navigate_to_slide	Jump to a page; persist current_page
get_slide_details	Narrate current or specified page from index
search_and_answer	Hybrid retrieve over slides; stay or navigate
mute_self / unmute_self	Hard silence path
leave_call	Explicit exit only
fetch_external_data	Last resort; not for on-deck questions

Spoken answers are instructed to come from **tool results** (slide_content / brief_content), not model memory.

8. Maturity

Solid enough to demonstrate	Explicit next (roadmap)
Join Meet via Recall; present deck as bot camera; realtime voice; tool-grounded slide Q&A; index quality gate	First-class escalate-to-human queue; stronger abstention at tool boundary; multi-instance WS fan-out; claim→source audit trail

9. Fit with a digital protégé product

Protégé concern	Overtone role
Join Meet / Zoom as participant	Meeting adapter (focus of current demos)
Follow conversation without guessing	Session + deck grounding + abstain
Respond when addressed	Realtime voice + tools
Escalate when unsure	Architectural fit — align with expert KB / human handoff

Overtone is best understood as a **live meeting adapter + grounding shell**, not a replacement for an expert-judgment knowledge system.

This document describes current engineering architecture as of the date above. Capabilities labeled roadmap are intentional design direction, not production guarantees.

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